

Question 1

Write a C program to store 'n' elements in an array using dynamic memory allocation and find the largest number.

Code

```
#include <stdio.h>

#include <stdlib.h>

int main() {

    int n, i;

    printf("Enter number of elements: ");

    scanf("%d", &n);


    int* a = (int*)malloc(n * sizeof(int));

    if (a == NULL) {

        printf("Memory allocation failed!\n");

        return 1;

    }

    printf("Enter elements:\n");

    for (i = 0; i < n; i++) {

        scanf("%d", &a[i]);

    }

    int max = a[0];

    for (i = 1; i < n; i++) {

        if (max < a[i]) {

            max = a[i];

        }

    }

    printf("Largest element: %d\n", max);

    free(a);

    return 0;

}
```

Sample Output

Enter number of elements: 4

Enter elements:

12

45

7

32

Largest element: 45

Question 2

Write a C program to print an array in reverse order using pointers.

Code

```
#include <stdio.h>

int main() {

    int n;

    printf("Enter size of array: ");

    scanf("%d", &n);

    int a[n];

    printf("Enter elements:\n");

    for (int i = 0; i < n; i++) {

        scanf("%d", &a[i]);

    }

    int *ptr = a + n - 1;

    printf("Array in reverse order:\n");

    for (int i = 0; i < n; i++) {

        printf("%d ", *ptr);

        ptr--;

    }

    printf("\n");

    return 0;

}
```

Sample Output

Enter size of array: 4

Enter elements:

10

20

30

40

Array in reverse order:

40 30 20 10

Question 3

Write a C program to store information of students using structure. Sort them by Name.

Code

```
#include <stdio.h>

#include <string.h>

struct Student {

    int roll;

    char name[50];

    float marks;

};

int main() {

    int n, i, j;

    printf("Enter number of students: ");

    scanf("%d", &n);

    struct Student s[n], temp;

    for (i = 0; i < n; i++) {

        printf("\nEnter details for student %d:\n", i + 1);

        printf("Roll number: ");

        scanf("%d", &s[i].roll);

        printf("Name: ");

        scanf("%s", s[i].name);

        printf("Marks: ");

        scanf("%f", &s[i].marks);

    }

    for (i = 0; i < n - 1; i++) {

        for (j = 0; j < n - i - 1; j++) {

            if (strcmp(s[j].name, s[j + 1].name) > 0) {

                temp = s[j];

                s[j] = s[j + 1];

                s[j + 1] = temp;

            }

        }

    }

}
```

```
printf("\n--- Student Details Sorted by Name ---\n");  
  
for (i = 0; i < n; i++) {  
    printf("Roll: %d | Name: %-10s | Marks: %.2f\n", s[i].roll, s[i].name, s[i].marks);  
}  
  
return 0;  
}
```

Sample Output

Enter number of students: 3

Enter details for student 1:

Roll number: 101

Name: Charlie

Marks: 85.5

Enter details for student 2:

Roll number: 102

Name: Alice

Marks: 92.0

Enter details for student 3:

Roll number: 103

Name: Bob

Marks: 78.4

--- Student Details Sorted by Name ---

Roll: 102 | Name: Alice | Marks: 92.00

Roll: 103 | Name: Bob | Marks: 78.40

Roll: 101 | Name: Charlie | Marks: 85.50

Question 4

Write a C program to add 'n' distances given in inch-feet system using structures.

Code

```
#include <stdio.h>

struct Distance {
    int feet;
    float inch;
};

int main() {
    int n, i;
    struct Distance d, sum = {0, 0.0};
    printf("Enter number of distances: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("\nEnter distance %d:\n", i + 1);
        printf("Feet: ");
        scanf("%d", &d.feet);
        printf("Inches: ");
        scanf("%f", &d.inch);
        sum.feet += d.feet;
        sum.inch += d.inch;
    }
    while (sum.inch >= 12.0) {
        sum.inch -= 12.0;
        sum.feet++;
    }
    printf("\nTotal Distance: %d feet %.2f inches\n", sum.feet, sum.inch);

    return 0;
}
```

Sample Output

Enter number of distances: 2

Enter distance 1:

Feet: 5

Inches: 8.5

Enter distance 2:

Feet: 3

Inches: 5.7

Total Distance: 9 feet 2.20 inches

Question 5

Write a C program to add two complex numbers by passing structures to a function.

Code

```
#include <stdio.h>

struct Complex {
    float real;
    float imag;
};

struct Complex addComplex(struct Complex c1, struct Complex c2) {
    struct Complex temp;
    temp.real = c1.real + c2.real;
    temp.imag = c1.imag + c2.imag;
    return temp;
}

int main() {
    struct Complex c1, c2, sum;
    printf("Enter real and imaginary parts for first complex number:\n");
    scanf("%f %f", &c1.real, &c1.imag);
    printf("Enter real and imaginary parts for second complex number:\n");
    scanf("%f %f", &c2.real, &c2.imag);
    sum = addComplex(c1, c2);
    printf("\nSum = %.2f + %.2fi\n", sum.real, sum.imag);
    return 0;
}
```

Sample Output

Enter real and imaginary parts for first complex number:

2.5 3.1

Enter real and imaginary parts for second complex number:

1.2 4.3

Sum = 3.70 + 7.40i

Question 6

Write a C program to maintain an inventory management system using structures.

Code

```
#include <stdio.h>

struct Item {
    int id;
    char name[50];
    int quantity;
    float price;
};

int main() {
    int n, i;
    printf("Enter number of inventory items: ");
    scanf("%d", &n);

    struct Item items[n];

    for (i = 0; i < n; i++) {
        printf("\nEnter details for item %d:\n", i + 1);
        printf("Item ID: ");
        scanf("%d", &items[i].id);
        printf("Item Name: ");
        scanf("%s", items[i].name);
        printf("Quantity: ");
        scanf("%d", &items[i].quantity);
        printf("Price per unit: ");
        scanf("%f", &items[i].price);
    }

    float grandTotal = 0.0;
    printf("\n--- Inventory Report ---\n");
    for (i = 0; i < n; i++) {
```

```
float itemTotal = items[i].quantity * items[i].price;

grandTotal += itemTotal;

printf("ID: %d | Name: %-10s | Qty: %d | Price: $%.2f | Total: $%.2f\n",
      items[i].id, items[i].name, items[i].quantity, items[i].price, itemTotal);
}

printf("Total Inventory Value: $%.2f\n", grandTotal);

return 0;
}
```

Sample Output

Enter number of inventory items: 2

Enter details for item 1:

Item ID: 101

Item Name: Keyboard

Quantity: 10

Price per unit: 25.50

Enter details for item 2:

Item ID: 102

Item Name: Mouse

Quantity: 15

Price per unit: 12.00

--- Inventory Report ---

ID: 101 | Name: Keyboard | Qty: 10 | Price: \$25.50 | Total: \$255.00

ID: 102 | Name: Mouse | Qty: 15 | Price: \$12.00 | Total: \$180.00

Total Inventory Value: \$435.00

Question 7

Write a C program for decimal to binary, octal, and hexadecimal conversion using manual conversion logic.

Code

```
#include <stdio.h>

void printBinary(int n) {
    if (n == 0) {
        printf("0");
        return;
    }
    int binary[32];
    int i = 0;
    while (n > 0) {
        binary[i] = n % 2;
        n /= 2;
        i++;
    }
    for (int j = i - 1; j >= 0; j--) {
        printf("%d", binary[j]);
    }
}

void printOctal(int n) {
    if (n == 0) {
        printf("0");
        return;
    }
    int octal[32];
    int i = 0;
    while (n > 0) {
        octal[i] = n % 8;
        n /= 8;
        i++;
    }
    for (int j = i - 1; j >= 0; j--) {
        printf("%d", octal[j]);
    }
}
```

```

    }
}

void printHexadecimal(int n) {
    if (n == 0) {
        printf("0");
        return;
    }
    char hex[32];
    int i = 0;
    while (n > 0) {
        int rem = n % 16;
        if (rem < 10) {
            hex[i] = rem + '0';
        } else {
            hex[i] = rem - 10 + 'A';
        }
        n /= 16;
        i++;
    }
    for (int j = i - 1; j >= 0; j--) {
        printf("%c", hex[j]);
    }
}

int main() {
    int num;

    printf("Enter a decimal number: ");
    scanf("%d", &num);

    printf("\nConversions for %d:\n", num);

    printf("Binary: ");
    printBinary(num);
    printf("\n");
}

```

```
printf("Octal: ");  
printOctal(num);  
printf("\n");  
printf("Hexadecimal: ");  
printHexadecimal(num);  
printf("\n");  
  
return 0;  
}
```

Sample Output

Enter a decimal number: 45

Conversions for 45:

Binary: 101101

Octal: 55

Hexadecimal: 2D

Question 8

Write a C program to calculate time period difference between two time periods.

Code

```
#include <stdio.h>

struct Time {
    int hours;
    int minutes;
    int seconds;
};

int main() {
    struct Time start, stop, diff;
    printf("Enter start time (hours minutes seconds): ");
    scanf("%d %d %d", &start.hours, &start.minutes, &start.seconds);
    printf("Enter stop time (hours minutes seconds): ");
    scanf("%d %d %d", &stop.hours, &stop.minutes, &stop.seconds);
    if (stop.seconds < start.seconds) {
        --stop.minutes;
        stop.seconds += 60;
    }
    diff.seconds = stop.seconds - start.seconds;
    if (stop.minutes < start.minutes) {
        --stop.hours;
        stop.minutes += 60;
    }
    diff.minutes = stop.minutes - start.minutes;
    diff.hours = stop.hours - start.hours;
    printf("\nTime Difference: %02d:%02d:%02d\n", diff.hours, diff.minutes, diff.seconds);
    return 0;
}
```

Sample Output

Enter start time (hours minutes seconds): 10 45 50

Enter stop time (hours minutes seconds): 13 20 15

Time Difference: 02:34:25

Question 9

Write a C program to implement an employee management system using structures.

Code

```
#include <stdio.h>

struct Employee {
    int id;
    char name[50];
    char designation[30];
    float salary;
};

int main() {
    int n, i;
    printf("Enter number of employees: ");
    scanf("%d", &n);
    struct Employee emp[n];
    for (i = 0; i < n; i++) {
        printf("\nEnter details for employee %d:\n", i + 1);
        printf("ID: ");
        scanf("%d", &emp[i].id);
        printf("Name: ");
        scanf("%s", emp[i].name);
        printf("Designation: ");
        scanf("%s", emp[i].designation);
        printf("Salary: ");
        scanf("%f", &emp[i].salary);
    }
    printf("\n--- Employee Records ---\n");
    for (i = 0; i < n; i++) {
        printf("ID: %d | Name: %-10s | Role: %-12s | Salary: $%.2f\n",
            emp[i].id, emp[i].name, emp[i].designation, emp[i].salary);
    }
    return 0;
}
```

Sample Output

Enter number of employees: 2

Enter details for employee 1:

ID: 501

Name: Alice

Designation: Manager

Salary: 75000

Enter details for employee 2:

ID: 502

Name: John

Designation: Developer

Salary: 60000

--- Employee Records ---

ID: 501	Name: Alice	Role: Manager	Salary: \$75000.00
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ID: 502	Name: John	Role: Developer	Salary: \$60000.00
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Question 10

Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

Code

```
#include <stdio.h>

#include <math.h>

float rectArea(float l, float b) { return l * b; }

float rectPerimeter(float l, float b) { return 2 * (l + b); }

float triArea(float a, float b, float c) {
    float s = (a + b + c) / 2.0;
    return sqrt(s * (s - a) * (s - b) * (s - c));
}

float triPerimeter(float a, float b, float c) { return a + b + c; }

int main() {
    float (*calc2Val)(float, float);
    float (*calc3Val)(float, float, float);
    float length = 5.0, breadth = 4.0;
    float side1 = 3.0, side2 = 4.0, side3 = 5.0;
    printf("--- Rectangle Calculations ---\n");
    calc2Val = rectArea;
    printf("Area: %.2f\n", calc2Val(length, breadth));
    calc2Val = rectPerimeter;
    printf("Perimeter: %.2f\n", calc2Val(length, breadth));
    printf("\n--- Triangle Calculations ---\n");
    calc3Val = triArea;
    printf("Area: %.2f\n", calc3Val(side1, side2, side3));
    calc3Val = triPerimeter;
    printf("Perimeter: %.2f\n", calc3Val(side1, side2, side3));
    return 0;
}
```

Sample Output

--- Rectangle Calculations ---

Area: 20.00

Perimeter: 18.00

--- Triangle Calculations ---

Area: 6.00

Perimeter: 12.00